msvtx_nTrk

MC vs data · overview · rank 12 / 98

rank 12 / 98 by endcap D_equal

used by: NN2 barrel, NN2 endcap

rank-identical columns merged into this one:
    MS1Vtx_nTracklet_raw
    MS1Vtx_barrel_nTracklet
    MS1Vtx_endcap_nTracklet

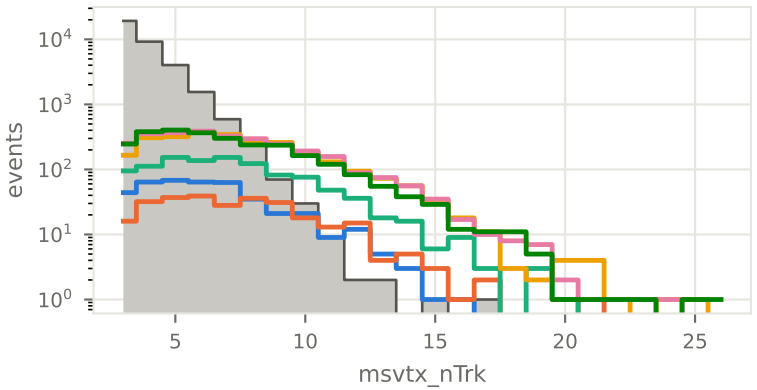
region	D_equal	noise	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg	rank
endcap	0.606	0.022	0.607	0.551	0.643	0.620	0.635	0.603	0.582	34,970	12/98
barrel	0.442	0.051	0.458	0.418	0.420	0.467	0.443	0.464	0.475	1,089	17/98
all	0.545	0.018	0.548	0.502	0.559	0.562	0.569	0.544	0.534	36,059	7/98

msvtx_nTrk

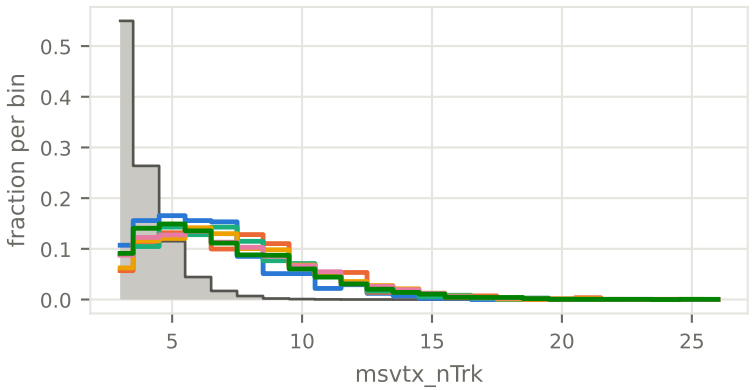
region: ENDCAP · rank 12 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.606 (noise 0.022) D_pooled 0.607 min 0.551 (mH125_mS15) max 0.643 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

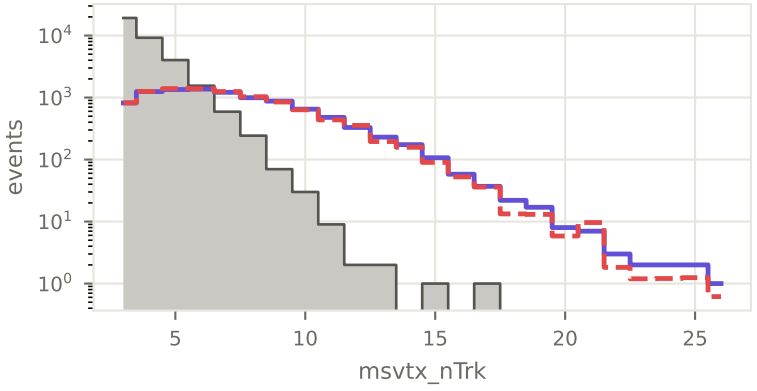


1b · each mass point vs background · normalized

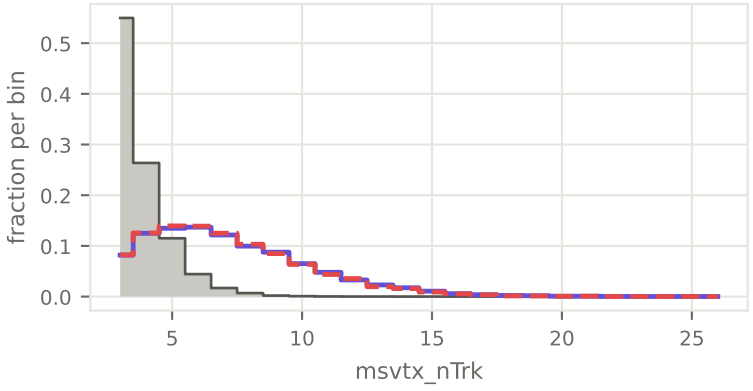


1 · each mass point
background n 34,970
mH125_mS15 D 0.551 n 411
mH125_mS55 D 0.643 n 281
mH200_mS50 D 0.620 n 1,071
mH400_mS100 D 0.635 n 2,648
mH600_mS150 D 0.603 n 2,882
mH1000_mS275 D 0.582 n 2,703

2a · combined signal vs background · raw

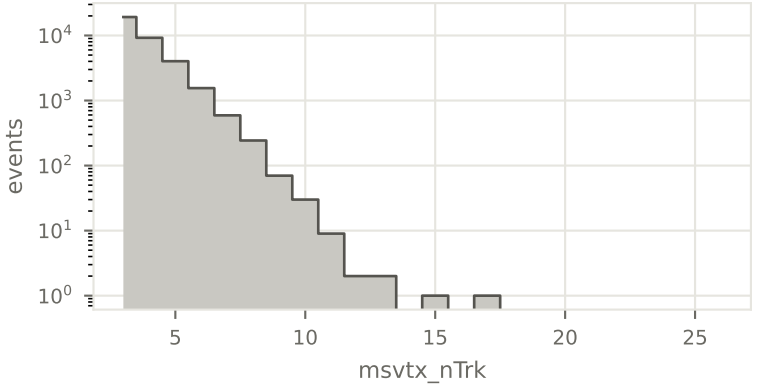


2b · combined signal vs background · normalized

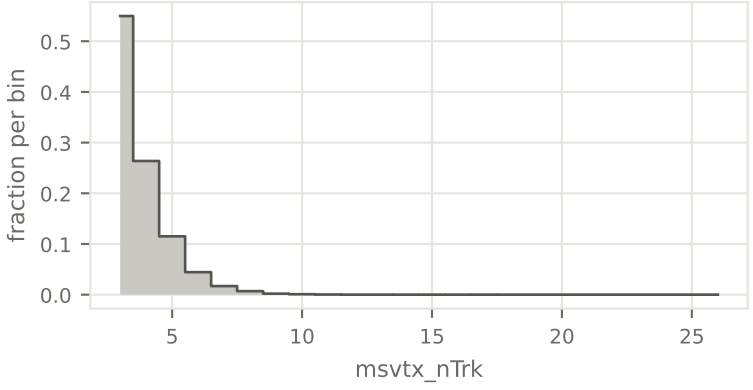


2 · combined signal
background n 34,970
combined, pooled D 0.607 n 9,996
equal weight per mass D 0.606

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

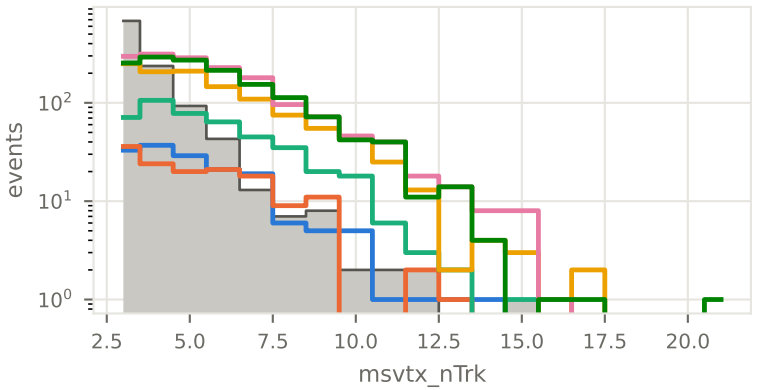
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nTrk

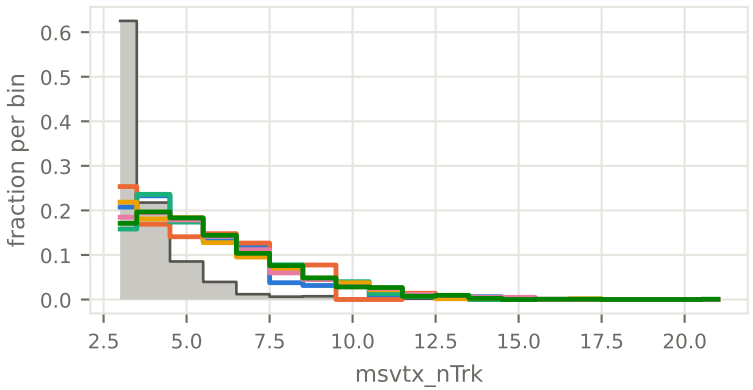
region: BARREL · rank 12 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.442 (noise 0.051) D_pooled 0.458 min 0.418 (mH125_mS15) max 0.475 (mH1000_mS275) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

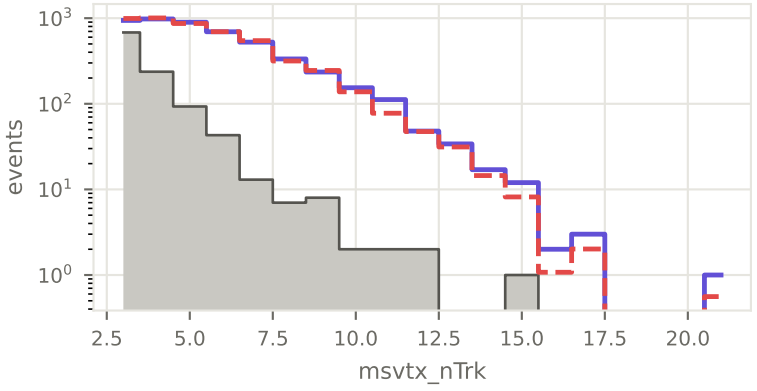


1b · each mass point vs background · normalized

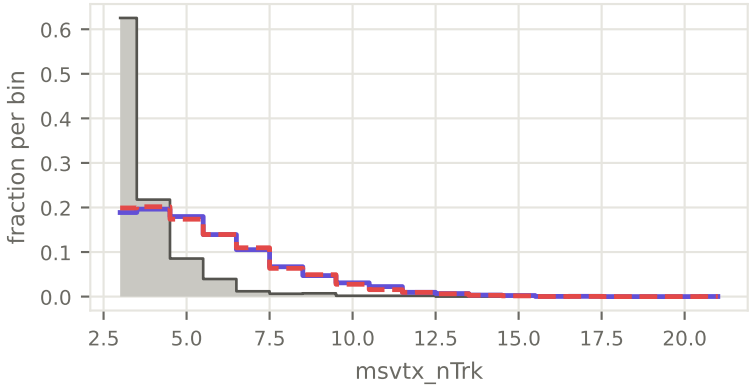


1 · each mass point
- background n 1,089
- mH125_mS15 D 0.418 n 159
- mH125_mS55 D 0.420 n 142
- mH200_mS50 D 0.467 n 449
- mH400_mS100 D 0.443 n 1,144
- mH600_mS150 D 0.464 n 1,608
- mH1000_mS275 D 0.475 n 1,485

2a · combined signal vs background · raw

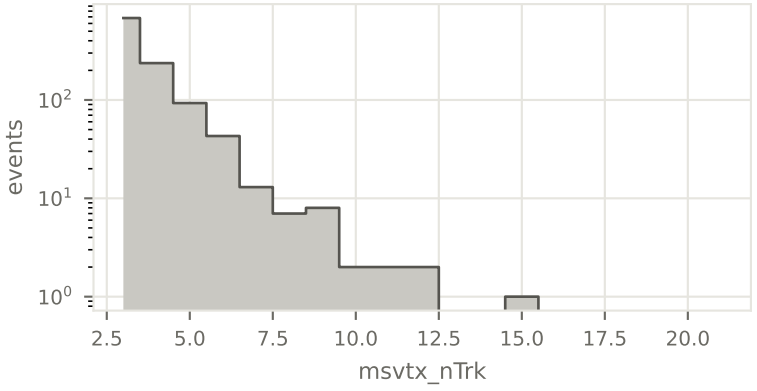


2b · combined signal vs background · normalized

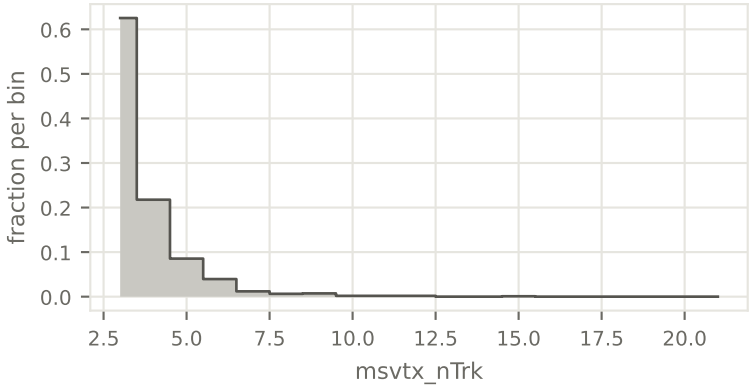


2 · combined signal
- background n 1,089
- combined, pooled D 0.458 n 4,987
- equal weight per mass D 0.442

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
- background n 1,089

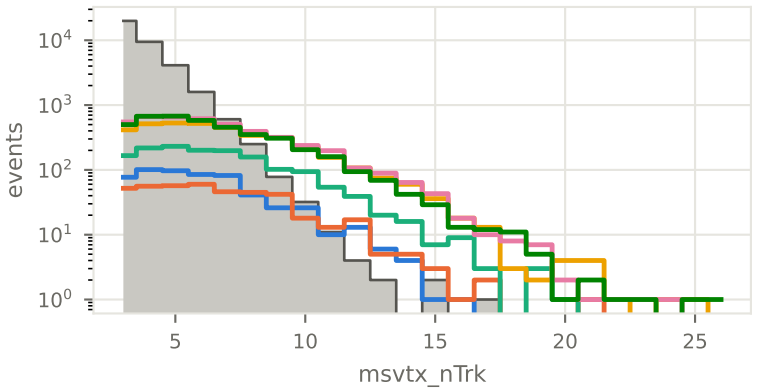
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_nTrk

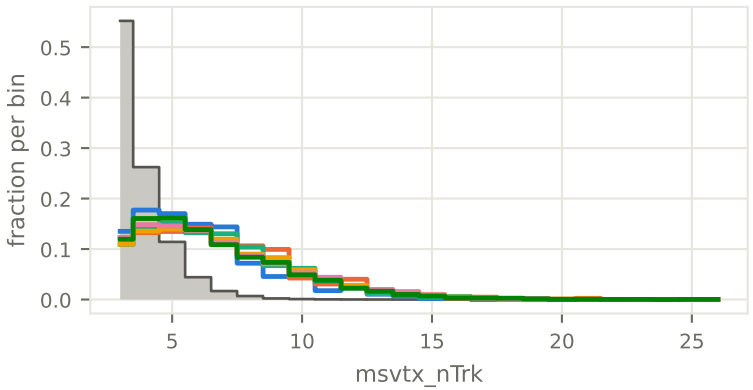
region: ALL · rank 12 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.545 (noise 0.018) D_pooled 0.548 min 0.502 (mH125_mS15) max 0.569 (mH400_mS100) bkg n = 36,059 sig n = 14,983

1a · each mass point vs background · raw

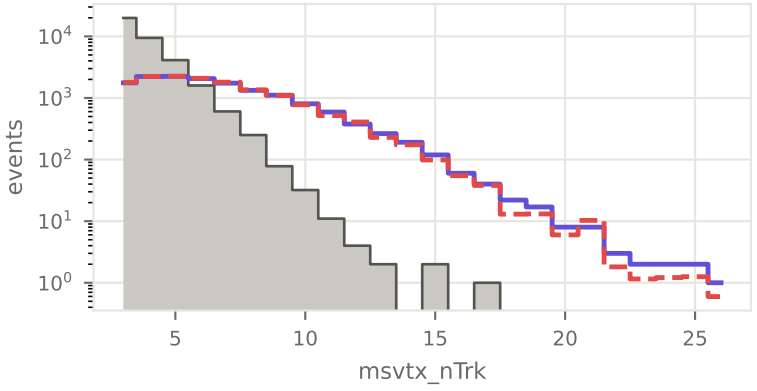


1b · each mass point vs background · normalized

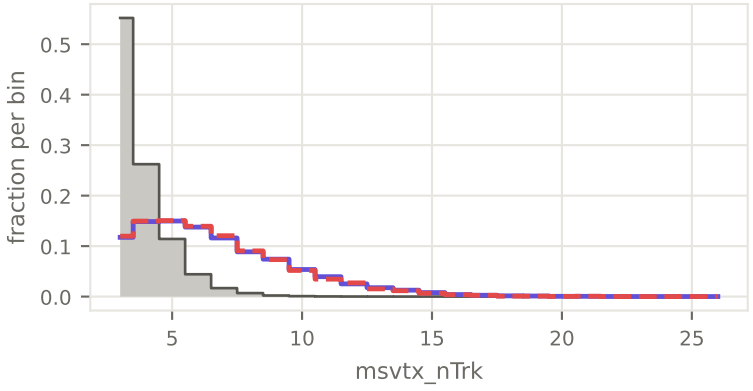


1 · each mass point
background n 36,059
mH125_mS15 D 0.502 n 570
mH125_mS55 D 0.559 n 423
mH200_mS50 D 0.562 n 1,520
mH400_mS100 D 0.569 n 3,792
mH600_mS150 D 0.544 n 4,490
mH1000_mS275 D 0.534 n 4,188

2a · combined signal vs background · raw

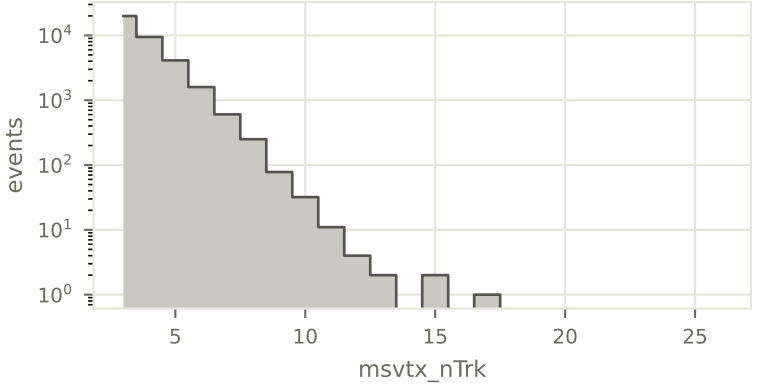


2b · combined signal vs background · normalized

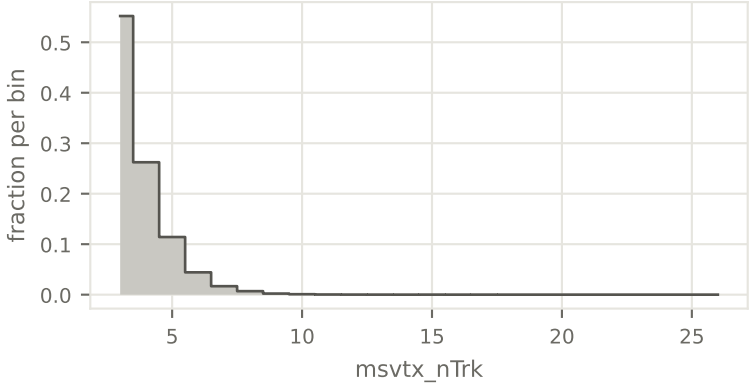


2 · combined signal
background n 36,059
combined, pooled D 0.548 n 14,983
equal weight per mass D 0.545

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 36,059

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed